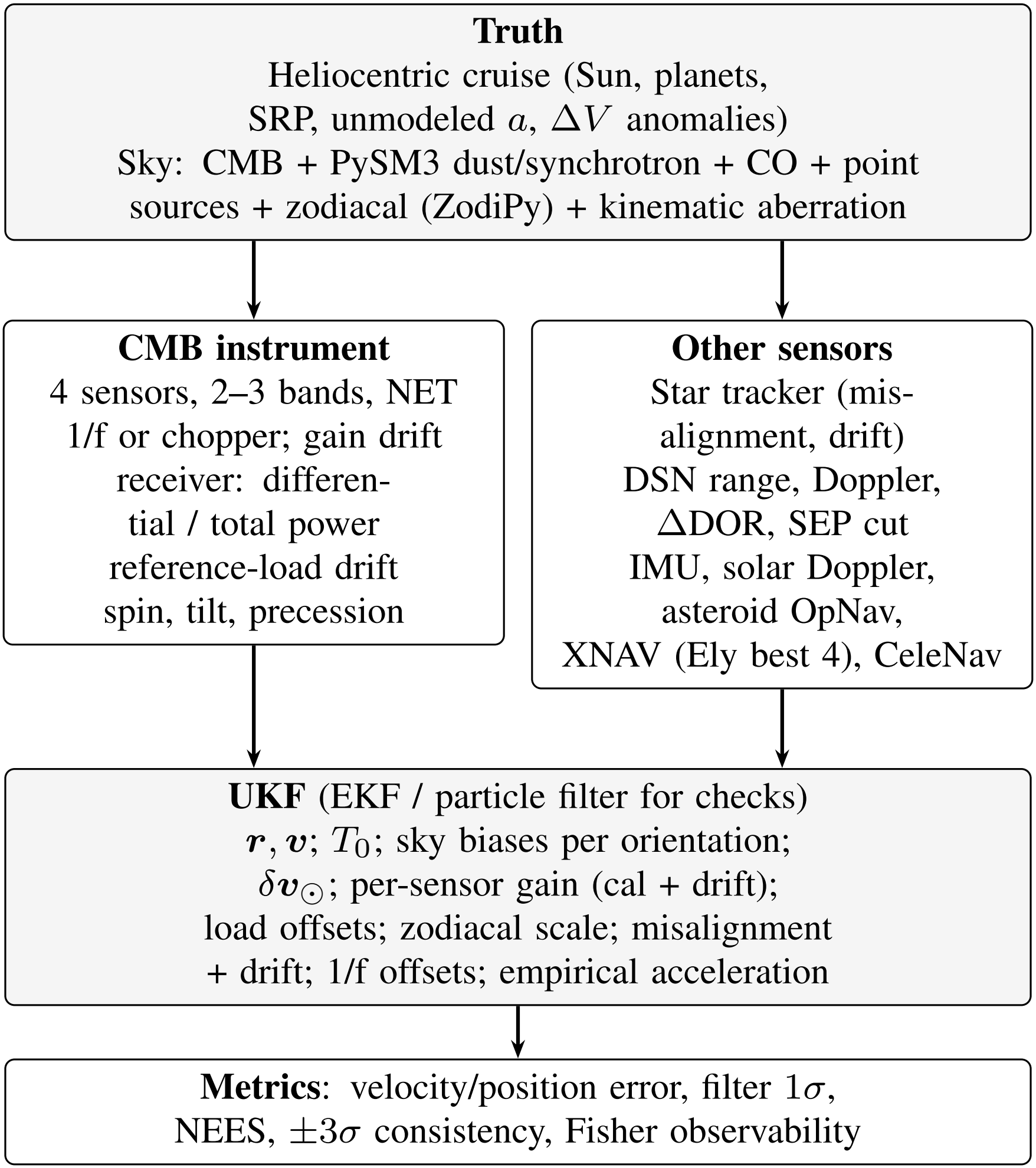

Truth
Heliocentric cruise (Sun, planets,
SRP, unmodeled a, ΔV anomalies)
Sky: CMB + PySM3 dust/synchrotron + CO + point
sources + zodiacal (ZodiPy) + kinematic aberration
CMB instrument
4 sensors, 2–3 bands, NET
1/f or chopper; gain drift
receiver: differen-
tial / total power
reference-load drift
spin, tilt, precession
Other sensors
Star tracker (mis-
alignment, drift)
DSN range, Doppler,
ΔDOR, SEP cut
IMU, solar Doppler,
asteroid OpNav,
XNAV (Ely best 4), CeleNav
UKF (EKF / particle filter for checks)
$\boldsymbol{r}, \boldsymbol{v}$; T_0; sky biases per orientation;
$\delta\boldsymbol{v}_\odot$; per-sensor gain (cal + drift);
load offsets; zodiacal scale; misalignment
+ drift; 1/f offsets; empirical acceleration
Metrics: velocity/position error, filter 1σ,
NEES, $\pm 3\sigma$ consistency, Fisher observability